

TABLE OF CONTENTS

E-301 OIL IMMERSED POWER TRANSFORMERS

	Page No.
1. SCOPE OF WORK	4
2. CODES AND STANDARDS	4
2.1 Indonesian National Standards	4
2.2 International Electrotechnical Commission	4
2.3 American National Standard Institute	6
2.4 Institute of Electrical and Electronics Engineers	6
2.5 National Electrical Manufacturers Association	6
3. OPERATING AND DESIGN REQUIREMENTS	7
3.1 Rated Power Output (MVA)	7
3.2 Vibrations and Noise Level	7
3.3 Overloads	7
3.4 Short Circuits	7
4. DETAILED REQUIREMENTS	8
4.1 Tank	8
4.2 Core and Coils	10
4.2.1 General	10
4.2.2 Transformer Coil Windings	10
4.2.3 Transformer Cores	11
4.3 Insulating Oil, Preservation Systems and Accessories	11
4.3.1 Insulating Oil	11
4.3.2 Oil Preservation Systems	12
4.3.3 Accessories and Fittings	14
4.4 Bushings and Current Transformers	17
4.4.1 Bushings	17
4.4.2 Current Transformers	18
4.5 Surge Arresters	19

4.6 On-Load Tap Changer	20
4.7 Cooling System	23
4.7.1 General	23
4.7.2 Motors	24
4.8 Control Cabinet, Devices and Wiring	25
4.8.1 General	25
4.8.2 Power Supply	25
4.8.3 Control and Lighting Devices	25
4.8.4 Cable Boxes and Glands	26
4.8.5 Cables and Wiring	26
4.8.6 Terminal Blocks	27
4.8.7 Fuses and Links	27
4.8.8 Motor Control Gear	27
4.8.9 Automatic Transfer Switch (If specified)	28
4.9 Photographs	28
4.10 Nameplates	29
4.11 Painting	29
4.12 Shipping.....	29
5. PERFORMANCE TESTING AND GUARANTEES	30
5.1 General	30
5.2 Factory Tests	31
5.2.1 Transformer Tests	31
5.2.2 Voltage Control Equipment	35
5.2.3 Cable Boxes and Disconnecting Chambers	35
5.2.4 Bushings	36
5.2.5 Tanks and ONAN Coolers	36
5.2.6 Cooling Plant with Forced Oil Circulation	37
5.2.7 Pump, Motors, Pipework, Oil Sampling Devices and Valves	37
5.2.8 Oil Tests	38
5.2.9 Gas and Oil Actuated Relays	38
5.2.10 Instrument Transformers	38
5.2.11 Insulation System	39
5.2.12 Control Cabinets, Devices and Wiring	39
5.3 Site Tests.....	39
5.4 Inspection and Witnessing of Tests	39

5.5 Test Reports.....	40
6. QUALITY ASSURANCE / QUALITY CONTROL	40
6.1 Requirement	40
6.2 Procedures	40
6.3 Documentation	41

E 301 OIL IMMERSED POWER TRANSFORMERS

1. SCOPE OF WORK

This Specification defines the general design requirement for large capacity outdoor oil-immersed power transformers.

2. CODES AND STANDARDS

The Contractor shall comply with the requirement of Part 4, Subsection 2.5 “Codes and Standards”. The following codes and standards to be included in this specifications are applicable to the design, manufacturing, and testing of equipment.

2.1 Indonesian National Standards

SPLN 61	:	Spesifikasi Transformator Tegangan Tinggi (Specification for High Voltage Transformer)
PUIL	:	Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)

2.2 International Electrotechnical Commission

IEC 60028	:	International standard of resistance for copper
IEC 60034	:	Rotating electrical machines
IEC 60044	:	Instrument transformers
IEC 60050	:	International Electrotechnical Vocabulary
IEC 60051	:	Direct acting indicating analogue electrical measuring instruments and their accessories
IEC 60059	:	IEC standard current ratings
IEC 60060	:	High-voltage test techniques
IEC 60071	:	Insulation co-ordination
IEC 60072	:	Dimensions and output series for rotating electrical machines
IEC 60073	:	Basic and safety principles for man-machine interface, marking and identification – Coding principles for indicators and actuators
IEC 60076	:	Power transformers
IEC 60137	:	Insulated bushings for alternating voltages above 1000 V
IEC 60156	:	Insulating liquids – Determination of the breakdown voltage at power frequency – Test method

IEC 60186	:	Voltage transformers
IEC 60214	:	Tap-changers
IEC 60228	:	Conductors of insulated cables
IEC 60245	:	Rubber insulated cables–Rated voltages up to and including 450/750 V
IEC 60255	:	Electrical relays
IEC 60269	:	Low-voltage fuses
IEC 60270	:	High-voltage test techniques – Partial discharge measurements
IEC 60282	:	High-voltage fuses
IEC 60296	:	Fluids for electrotechnical applications – Unused mineral insulating oils for transformers and switch-gear
IEC 60309	:	Plugs, socket-outlets and couplers for industrial purposes
IEC 60331	:	Test for electric cables under fire conditions – Circuit integrity
IEC 60332	:	Tests on electric cables under fire conditions
IEC 60354	:	Loading guide for oil-immersed power transformers
IEC 60422	:	Supervision and maintenance guide for mineral insulating oils in electrical equipment
IEC 60423	:	Conduits for electrical purposes – Outside diameters of conduits for electrical installations and threads for conduits and fittings
IEC 60437	:	Radio interference test on high-voltage insulators
IEC 60445	:	Basic and safety principles for man-machine interface, marking and identification – Identification of equipment terminals and of terminations of certain designated conductors, including general rules for an alphanumeric system
IEC 60446	:	Basic and safety principles for man-machines interface, marking and identification – Identification of conductors by colours or numerals
IEC 60475	:	Method of sampling liquid dielectrics
IEC 60502	:	Power cables with extruded insulation and their accessories for rated voltages from 1 kV ($U_m = 1.2$ kV) up to 30 kV ($U_m = 36$ kV)
IEC 60507	:	Artificial pollution tests on high-voltage insulators to

		be used on a.c. systems
IEC 60529	:	Degrees of protection provided by enclosures (IP Code)
IEC 60542	:	Application guide for on-load tap-changer
IEC 60567	:	Guide for the sampling of gases and of oil from oil-filled electrical equipment and for the analysis of free and dissolved gases
IEC 60590	:	Determination of the aromatic hydrocarbon content of new mineral insulating oils
IEC 60599	:	Mineral oil-impregnated electrical equipment in service – Guide to the interpretation of dissolved and free gases analysis
IEC 60605	:	Equipment reliability testing
IEC 60614	:	Conduits for electrical installations – Specification
IEC 60616	:	Terminal and tapping markings for power transformers
IEC 60617	:	Graphical symbols for diagrams
IEC 60811	:	Common test methods for insulating and sheathing materials of electric cables and optical cables
IEC 60947	:	Low-voltage switchgear and controlgear
IEC 61082	:	Preparation of documents used in electrotechnology
IEC 61125	:	Unused hydrocarbon based insulating liquids – Test methods for evaluating the oxidation stability
IEC 62155	:	Hollow pressurized and unpressurized ceramic and glass insulators for use in electrical equipment with rated voltages greater than 1000 V
IEC 62271	:	High-voltage switchgear and controlgear

2.3 American National Standard Institute

ANSI C 57.98 : IEEE Guide for Transformer Impulse Tests

2.4 Institute of Electrical and Electronics Engineers

ANSI/IEEE 454 : Recommended practice for the detection and measurement of partial discharges (corona) during dielectric tests

2.5 National Electrical Manufacturers Association

NEMA 107 : Methods of measurement of Radio Influence Voltage

(RIV) of high-voltage apparatus

3. OPERATING AND DESIGN REQUIREMENTS

The transformer shall be suitable for operation as specified at a tropical site, refer to Part 4, Subsection 1.1, 2.14 and 2.16 for meteorological, seismic and environmental conditions at the site.

Generally all hardware and accessories shall be corrosion proof materials.

3.1 Rated Power Output (MVA)

Each transformer shall have a rated power output (MVA) in accordance with its detailed specifications and shall be capable to operate without exceeding the specified temperature rise under site conditions, as follows:

- (1) Delivering its rated power from any winding continuously, irrespective of the direction of power flow, at any tapping, with the output winding at $\pm 5\%$ rated voltage, the frequency at $\pm 2\%$ rated and a load power factor between 0.8 lagging and 0.9 leading.
- (2) Operating continuously at no load at any tapping, when neither the output winding voltage nor Volts per Hertz exceeds 110% rated.

For multi-winding transformer each winding shall be capable of being loaded to their specified ratings irrespective of the loading on other windings, but the arithmetic sum of the winding loads shall not exceed the source winding capability.

3.2 Vibrations and Noise Level

The maximum vibration level of the transformers and its appurtenances shall be such that visible motion should not be detectable.

The maximum noise level for the power transformers shall be in accordance with Part 4, Section 2, Subsection 2.9 for Noise Level Requirement.

3.3 Overloads

Each transformer and all component parts shall be capable of overload operation in accordance with IEC 60354.

3.4 Short Circuits

The transformer shall be capable of withstanding, without any permanent

damage or deterioration, the mechanical and thermal stresses caused by short circuits on the external terminals of any winding or windings in accordance with IEC 60076.

4. DETAILED REQUIREMENTS

4.1 Tank

The transformer tank shall be of welded steel construction capable of withstanding 125% of the maximum positive and negative operating pressures as a minimum. A minimum of one manhole of the size indicated (or maximum size possible) shall be provided for each transformer. For transformer above 12.5 MVA three-phase there shall be more than one manhole in the transformers cover to facilitate the removal and installation of bushings and current transformers, tap changer mechanisms, winding temperature current transformers, etc., and to provide access to the terminal board without disturbing the leads. Manholes, if circular, shall be a minimum of 600 mm in diameter. If rectangular or oval, they shall have a minimum dimension of 600 x 450 mm.

All joints of oil-filled fabricated chambers, other than those which have to be broken, shall be welded and care shall be taken to ensure that the chambers are oil-tight. Defective welded joints shall not be caulked but may be re-welded.

Oil filling level indication shall be engraved on the inside and outside of chambers.

All joint faces shall be machined flat and parallel and arranged to prevent the ingress of water or leakage of oil with a minimum of gasket surface exposed to the action of oil or air.

The tank radiator and conservator (if required) shall be designed to withstand vacuum filling at full vacuum in the field.

Special precautions shall be taken to ensure that any gas evolved is directed as quickly as possible to the gas detector relay. The tank covers shall be slanted or domed away from the direction of the major axes to prevent gas from being trapped at the ends or sides of the tank cover. The gas collection

point shall be located at the “high spot” of the cover. If there are any internal pockets where gas could collect they shall have a collar extending down into the oil, so that only that gas which is evolved vertically below the collar is trapped by the collar. All such pockets shall be piped to the gas detector relay and shall also be provided with bleed devices for venting of large quantities (over 800 cm³) of trapped air or gas. Any gas collection pockets which cannot be piped to the gas detector relay; such as the pressure relief vent pipe and spaces between bushing draw leads and bushing central tubes, shall be provided with gas deflectors to prevent gas from entering such points.

The tank shall have a steel skid type under base, a minimum of 20 mm thick to permit moving the transformer in any direction.

Pulling eyes shall be provided near the tank base and shall be located to permit skidding and slewing the transformer through an angle of 90°. The pulling eyes shall each be suitably braced to withstand a pull in any direction at up to 15° from the horizontal, when the transformer is completely assembled and filled with oil.

Four jacking steps shall be provided in the line with the foundation pier center-lines and shall have a minimum height of 350 mm. Each jacking step shall have a capacity for lifting one half the weight of the completely assembled transformer with oil.

The tank cover shall be designed to shed water and shall be bolted to the main tank. Lifting eyes shall be provided to facilitate removal. Two full capacity ground connections shall be provided between the cover and the tank.

All pipe entries to the main tank shall have oil-tight shut-off valves, which can be locked in either the open or closed position and arranged to give positive indication for both positions.

The tank shall be complete with filling and drain valves with capped male fitting. The drain valve cap shall incorporate an oil sampling connection and valve. Drain holes shall be provided in all box stiffeners.

All metallic parts, exclusive of those energized, shall be grounded.

4.2 Core and Coils

4.2.1 General

Cores and coil shall be braced to withstand the short circuit forces without damage or displacement of the coil on the core, and to withstand normal moving and handling without the use of special shipping braces.

Suitable cooling liquid passages shall be provided to limit the hot spot temperature rise above the average winding temperature rise at rated load to those values specified in IEC 60076.

Lifting lugs shall be included to facilitate removal of the complete core and coil assembly for repairs.

The current carrying capability of the transformer shall be limited only by the capacity of the core and coils, not by the capacity of other current carrying components such as bushings and tap changers.

4.2.2 Transformer Coil Windings

Transformer coil windings shall be fabricated of copper conductor.

The coils shall be insulated from the core and each other with sufficient insulation to withstand the specified impulse and low frequency dielectric tests for transformers of the specified voltage class.

Each transformer shall be designed so that with the specified impulse test voltage or the maximum “let-through” voltage of the associated surge arresters, applied to any winding, the maximum surge that can be transferred to the other windings, shall not exceed their specified insulation level.

Tertiary or Stabilizing Windings

- (1) Where tertiary windings are required they shall be provided by the Contractor. They shall be designed to withstand without injury the effects of short circuits on the tertiary or other windings up to the specified short circuit capabilities.
- (2) Where tertiary windings are not specified, but are required, a stabi-

lizing winding or “buried” tertiary can be provided.

- (3) On three phase units one corner of the “buried” or stabilizing winding shall be provided with two removable test links, one to open the delta and the other to open the ground connection. These links shall be located inside the transformer tank and shall be easily accessible through a manhole in the tank cover.

4.2.3 Transformer Cores

Frame and core bolts shall be insulated from the core.

The cores, framework and clamping arrangements shall be capable of withstanding any shocks to which the transformers may be subjected. Movement of the transformer core relative to the tank shall be prevented.

Care shall be taken to maintain even mechanical pressure over the whole of the laminated core. Each lamination shall be insulated with a material which will not deteriorate under pressure and the action of hot oil.

The transformer shall be designed to avoid undesirable harmonics, especially third and fifth harmonics.

The core ground connection shall be accessible from a manhole without removing any oil from the transformer tank or climbing into the tank.

4.3 Insulating Oil, Preservation Systems and Accessories

4.3.1 Insulating Oil

The transformer shall be provided with the necessary amount of high grade mineral insulating oil, Shell Diala Oil B. The oil shall be in accordance with the IEC 60296 (Class 1) Standard.

Where possible, the transformer shall be shipped filled to the 25 °C oil level. If the transformer cannot be shipped filled with oil it shall be filled with nitrogen or dry air in accordance with the manufacture’s proven standard procedures.

If the dielectric test strength of any oil received in the field, whether shipped separately or in the transformer, tests less than 30 kV/2.5mm when tested in

accordance with the procedures of IEC 60156.

When oil samples taken from the completely filled transformer and tested in accordance with IEC 60156, the samples shall have dielectric test strength of 26 kV/2.5 mm.

4.3.2 Oil Preservation Systems

Transformers shall have an oil preservation system for protecting the oil against atmospheric moisture and oxygen. The oil preservation system of each transformer shall be the manufacture's standard. Inert gas systems, "sealed tank" systems or conservator systems shall be conformed to their respective requirements.

Inert Gas System

- (1) Each automatically maintained inert gas pressure system shall provide a dry nitrogen gas atmosphere above the oil. The complete gas system shall include separate high-pressure nitrogen tanks, automatic pressure regulating valve, high and low pressure gauges and alarm devices with alarm-contacts arranged for remote indication of low gas cylinder pressure and low transformer gas pressure.
- (2) Valves shall be provided to permit purging the gas space and testing the seal on the tank by admitting dry nitrogen under pressure. A gas-sampling valve shall be located opposite the nitrogen gas tank inlet.
- (3) There shall also be included an adequate amount of bottled nitrogen gas to place the transformer in service and for all required testing.

Sealed Tank System

- (1) All handhole covers and bushings shall be securely bolted to gasket openings in the cover. The cover shall be welded in place. Each preservation system shall also include a pressure vacuum gauge and a pressure relief device designed to seal the interior of the transformer from the atmosphere and hold the gas plus oil volume constant without the internal gas pressure exceeding 0.70 kg/cm² (10 psi) gauge positive or 0.56 kg/cm² (8 psi) gauge negative.
- (2) Sufficient expansion head shall be provided to allow an ambient temperature range from 10°C to 50°C under all design and load conditions without exceeding pressure limits or operating pressure-vacuum bleeder device.

- (3) The transformer shall be shipped with all radiators, bushings, covers, and connection in place, ready to unload, install and place in immediate service.

Oil Conservator Systems

- (1) Oil conservator type preservation system shall be complete with filling cap, sump and drain valve and having adequate capacity for the ambient temperature range at the site.
- (2) The capacity between the highest and lowest visible levels shall not be less than 7.5% of the total cold oil volume in the transformer and cooling equipment.
- (3) The tank shall be mounted with 25 mm slope between the ends to facilitate draining and the lower end shall be fitted with 25 mm drain/filter pressure valve complete with dummy plug. The conservator shall be provided with a manhole, 50 mm filling valve with dummy plug and lifting eyes. The connecting pipe to the main transformer tank shall be provided with an indicating shut-off valve located on the conservator side of the gas detector relay. The shut-off valve shall have a lever to indicate “valve open” when in line with the pipe and “valve closed” when at right angles to the pipe. The “open” and “closed” positions of the handle shall be easily observable from ground level. The handle shall be arranged so that it can be attached in only one way.
- (4) The main oil feed pipe from the conservator to the transformer shall be connected to the highest point of the transformer tank and shall be arranged at a rising angle towards the conservator of from 3 to 7 degrees to the horizontal and shall have a minimum internal diameter of 75 mm. Sharp bends in the pipe work shall be avoided.
- (5) The oil preservation system shall be such that there is no continuous contact between the oil in the conservator and the atmosphere. A dehydrating breather containing a color indicator shall be piped to the conservator and mounted at eye level.
- (6) A dehydrating type breather, by itself, is not acceptable, either of the following methods may be used.
 - a. Diaphragm (Air Cell) Type
Conservator with an oil resistant diaphragm (air cell) as an interface between the oil and the air. A dehydrating breather shall be provided on the

pipe that vents the air in the conservator to the atmosphere.

b. Automatic Breather

The automatic breather shall be of the thermal cycle type with adjustable refrigerating and defrosting periods to form a continuously repeated cycle and shall be completed with a cyclometer to record the number of frosting and defrosting cycles.

The automatic breather shall be mounted on the conservator. The automatic breather shall extract moisture from the air above the oil in the conservator. The water outlet from the automatic breather shall be piped to a measuring vessel at ground level and both the measuring vessel and pipe shall be of stainless steel or other approved material that is not subject to corrosion. The automatic breather's air intake from the atmosphere shall be via a dehydrating breather of ample capacity.

The automatic breather shall include all automatic and auxiliary control equipment which shall be accessible from ground level. The equipment alarm contacts shall be wired to the control cabinet.

4.3.3 Accessories and Fittings

All equipment requiring frequent inspection shall be installed in easily accessible location.

All indicating thermometers, oil level gauges, etc. shall be located so they can be easily read from ground level.

Electrical Contact Ratings

All electrical contacts provided for alarming or tripping shall be suitable for 250 V DC ungrounded service, with a minimum of 0.5 ampere, inductive rating.

Gas Detection Device

- (1) A gas detector relay shall be provided only if expansion head (conservator) equipment is furnished. The device shall be a two element Bucholz type and shall have a magnetically coupled and float operated gauge mounted on a float chamber in the pipe between the transformer and the conservator. It

shall be arranged to collect all gas evolved.

- (2) The relay shall be located for optimum viewing of the sight glass without de-energizing the transformer. A copper tube, minimum 10 mm inside diameter, shall extend from the relay to a gas sampling petcock mounted on the side of the transformer tank and accessible from ground level. The petcock shall be suitable for attaching a rubber tube of 6 mm inside diameter.
- (3) The relay shall be equipped with two sets of contacts; one set to operate an alarm on gas accumulation and one set to trip the transformer on oil surge.
- (4) Individual DC auxiliary seal-in relays shall be furnished with each pressure device. The auxiliary relays shall each have two normally open contacts. A reset pushbutton shall be provided. The Contractor shall provide separate DC tripping supplies for each relay.

Fault Pressure Device

- (1) A device shall be provided in each transformer and load tap-changer to close a contact for relay operation in case the internal pressure rises suddenly.
- (2) Individual DC auxiliary seal-in relays shall be furnished with each pressure device. Each auxiliary relay shall have two normally open contracts. A reset pushbutton shall be provided. The Contractor shall provide separate DC tripping supplies for each relay.

Pressure Relief Device

A pressure relief device shall be furnished for transformers above 2.5 MVA or 200 kV BIL for relieving excessive pressure within the transformer tank. Alarm contacts shall close on operation of the device. The device shall be automatic reset type with mechanical indicator (contacts to be manual reset). The device shall be designed to maintain the oil tightness of the transformer under all conditions of normal service.

Standard Accessories and Fittings

Each transformer shall be provided with a complete set of manufacture's standard accessories and fittings which as a minimum shall include :

- (1) Oil Level Gauges
 - a. A magnetic oil level gauge mounted on the wall of the transformer tank,

complete with low level alarm/trip contacts. The dial shall indicate when the oil is above the core and windings during filling operations.

- b. A magnetic oil level gauge, complete with low level alarm contacts, shall be mounted on the end of the conservator tank (if provided). The dial shall indicate the minimum, maximum and normal 25°C levels.

(2) Temperature Indicators

- a. Due to seismic design requirements, mercury switches will not be acceptable.
- b. Winding and oil temperature indicators shall be supplied as described below:
 - b.1 Dial type winding temperature indicator, responsive to the combination of top oil temperature and winding current, calibrate to follow the hottest spot temperature of the transformer winding. The dial shall be mounted at eye level. The temperature indicator shall be provided with separately adjustable sets of contacts to perform the following functions:
 - b.1.1 Initiate the first stage of forced cooling when the temperature reaches 75°C.
 - b.1.2 Initiate the second stage of forced cooling when the temperature reaches 80°C.
 - b.1.3 Operate an alarm when the temperature reaches 105°C.
 - b.1.4 Initiate a trip of the transformer load side breaker(s) when the temperature reaches 120°C.
- c. The winding temperature indicator shall have provision for field calibration.
- d. Dial type, oil temperature indicator with its sensing element located in the path of the hottest oil. The dial shall be mounted adjacent to that of the winding temperature indicator. The temperature indicator shall be equipped with separately adjustable sets of contacts to perform the following functions :
 - d.1 Operate an alarm when the temperature reaches 90°C.
 - d.2 Initiate a trip of the transformer load side breaker(s) when the temperature reaches 105°C.

- e. Resistance Temperature-element Device (RTD) for measuring the hot spot temperature of each winding, with external connections.

(3) Valves

The following valves shall be provided on the main transformer tank:

- a. 50 mm drain valve (minimum)
- b. 12 mm sampling valve
- c. 38 mm upper filter pressure valve
- d. 38 mm lower filter pressure valve mounted on the opposite side of the tank diagonally opposite the upper filter pressure valve
- e. 50 mm vacuum pulling valve on tank cover
- f. 50 mm vacuum oil spray valve on tank cover

All valves shall be gate type and shall be provided with dummy plugs or blanking plates.

Grounding Terminal Pads

Two (2) sets grounding terminal pads, mounted diagonally opposite each other and near the bottom of the transformer. Similar ground pad(s) shall also be provided on the cover of three phase transformers near the neutral bushing(s). For solidly grounded transformer neutrals, suitable copper ground connections shall be provided from the neutral bushing to the ground pad on the top of the tank.

Ladder

A metal ladder shall be attached to the side of the transformer tank and conveniently located to allow easy access to the gas detector relay and the top of the tank.

4.4 Bushings and Current Transformers

4.4.1 Bushings

All bushings shall be outdoor type and shall comply with the requirements of IEC 60137. The associated barrel porcelains shall be comply with IEC 62155 together with the requirements of this specification. Lifting lugs shall be included.

Transformer bushings for 33 kV and above shall be filled with transformer oil and there shall be no connection with the oil in the transformer. Bushings above 110

kV shall be the condenser type. An oil gauge readable from ground level shall be provided. The visible oil levels in the gauge shall correspond to average oil temperatures from the minimum ambient stated in the specifications to plus 70°C. The oil level at 15°C shall be marked.

Connections from the main windings to bushings shall be flexible and shall be such that undue mechanical stresses are not imposed on them during assembly on site. On oil filled bushings, the metal distributing plate shall terminate under oil. Oil filled bushings shall have power factor test plugs.

All current carrying surface of power transformer bushings, except for the terminal adapters, shall be copper or copper alloy material and shall be plated to inhibit deterioration. The terminal adapter pad may be aluminium provided proper care is taken in making the copper to aluminium connections.

4.4.2 Current Transformers

All current transformers shall comply with IEC 60044 and this Technical Specifications.

Current transformers shall be bushing type with fully distributed windings, a relay/metering accuracy Class 5 P 20/0.2 on all taps and minimum of 100 VA capacity, unless indicated otherwise.

All current transformer leads, including those from each tap of multiple ratio bushing current transformer, shall be terminated in IPW 66 terminal boxes on the exterior of the transformer tank and shall be extended into the transformer control cabinet.

All bushing type current transformers shall be removable without removing the transformer cover.

Current transformers shall be capable of withstanding, without damage, the peak and rated short-time currents of their associated equipment. Standard secondary current shall be 5 amperes with an extended current rating of 200%, unless otherwise specified or approved by the Owner.

The secondary windings of each set of current transformers shall be grounded at one point only via a bolted disconnecting link. The current transformer core,

where accessible, shall be separately grounded.

Where tapped secondary windings are specified a label shall be provided at the secondary terminals of the current transformer indicating clearly the connections required for each tap. The connections and the ratio in use shall be indicated on all connection diagrams.

Design magnetization curves and DC resistance values shall be submitted for each current transformer to be used for protective relaying purposes and shall subsequently verified by routine factory tests and test certificates.

The current transformer burden capability shall be the manufacturer's largest available rating for each ratio.

4.5 Surge Arresters

Surge Arresters of the "station" type shall be furnished for each bushing terminal where specified. The surge arresters shall be the metal oxide type not susceptible to failure because of external contamination. Refer to Specification E-430 Surge Arresters.

Each arrester shall be self-supporting and designed for base mounting.

Corona free connections between surge arresters and bushings shall be provided unless otherwise specified. These connections shall have a flexible arrangement, so that the bushing will not be overstressed during vacuum filling

All ferrous parts exposed to the weather and subject to corrosion shall be zinc coated by the hot dipped galvanized process.

Each arrester shall have a pressure relief device which will act positively to vent gas pressure for prevention of explosion of the porcelain in the event of arrester failure.

The color of the arresters shall be chocolate brown, unless otherwise specified.

An automatic surge arrester discharge counters shall be provided with each surge arrester stack. The discharge counter shall be complete with arrester insulating sub-base and hanger hardware. Discharge counter shall be equipped with a leak-

age current meter.

Contractor shall mount discharge counters on transformers and shall furnish two (2) 25 x 7 mm copper bars from the base on each surge arrester to the discharge counter and then to the ground pad at the base of the transformer.

4.6 On-Load Tap Changer

The transformer shall be provided with an on-load tap changer.

The on-load tap changer shall comply with all applicable portions of the latest issue of IEC 60214 unless otherwise specified.

The on-load tap changer shall be provided with all necessary equipment mounted at the transformer for manual and automatic operation. The equipment shall include a manual “Lower-Raise” control switch and rotating synchronous flush mounted tap position indicators for local control at the transformer.

The on-load taps shall be controlled by motor operated equipment of proven reliability and requiring a minimum of maintenance.

Overload and short-circuit characteristics shall be compatible with those of the transformer. All taps shall be rated for full capacity.

All on-load tap changer components shall be designed for the full continuous rating of the transformer. The drive mechanism shall be designed so that it is impossible for the tap changer to stop in any non-fully rated position. This shall apply whether the tap changer is driven by the motor or by manual hand operation.

The tap changer components shall start on a “Raise” or “Lower” impulse of short duration and shall “seal-in” and complete the step by itself. The duration of a signal beyond the time required for operation shall not initiate a second operation. In the event of power failure during a tap change operation, the tap changer shall automatically complete its operation to the next upper or lower tap position as soon as power returns. The return of power shall not initiate a second operation.

Electrical limit switches and mechanical stops or de-clutching devices shall be incorporated to prevent over-travel beyond the maximum raise and lower positions. Electrical and mechanical interlocks shall be provided on the “Raise” and “Low-

er” contactors.

Tap changer motor protection shall consist of an overcurrent automatic air circuit breaker with manual reset, or alternatively, a contactor with an isolating device, fuses and single phasing protection. An alarm shall be provided for operation of any motor protective devices.

The diverter switches shall be located in separate compartment so that the oil is completely isolated from the main transformer oil. It shall be possible to drain and sample the oil from this compartment for maintenance purposes without lowering the oil in the main transformer compartment.

Each tap changer oil-filled compartment shall be provided with a magnetic oil level gauge with low level alarm contacts, pressure relief device, filling/filter pressure valve and drain/filter pressure valve.

The diverter switch compartment shall also be provided with a fault pressure relay with auxiliary contacts or, alternatively, auxiliary contacts may be provided on the pressure relief device. A dehydrating type breather shall be provided and mounted at eye level. The breather shall contain a color indicator and shall be of a type in which the desiccant is not continuous contact with the atmosphere.

When the design of tap changer requires an oil expansion tank, it shall be piped to the main transformer conservator. It shall preferably be piped to a separate compartment in the conservator although it may be piped to the transformer compartment if an approved filter is provided in the tap changer to conservator oil connection pipe. If piped to separate compartment, the compartment shall be provided with its own magnetic oil level gauge with low level alarm contacts, filling/filter pressure valve and drain valve. A fault pressure relay with auxiliary contacts shall be provided in the pipe to the conservator. A shut-off valve shall be provided in the conservator pipe, on the conservator side of the fault pressure relay.

All oil level gauges shall be legible from ground level. Filling and drain valves shall be not less than 25 mm complete with dummy plugs.

The tap changer voltage regulator shall be designed to maintain low voltage winding(s) at or above the selected set point.

The regulator shall be designed such that the taps will not be changed unless the voltage deviation is greater than a preset value. The deviation shall be adjustable from $0 \pm 5\%$ of the rated winding voltage. Other adjustments and controls shall be the manufacturer's standard.

Line drop compensation shall be provided when specified and shall be of the resistance and reactance type. Current and potential test switches shall be provided for test purposes. Blocking facilities shall be provided to prevent the tap changer from raising if the voltage rises above an adjustable pre-set value and from lowering if the voltage falls below an adjustable pre-set value.

Facilities shall be provided when specified, for transformer paralleling controls.

Remote controls shall be provided with the necessary equipment to provide for remote tap changer control, including a key lock "Local-Remote" selector switch with the key removable in both positions, a digital type tap position indicator, "Raise/Lower" push buttons or a selector switch, a "tap change in progress" indication lamp and buzzer for each transformer. The equipment shall be mounted and wired on a control panel or for installation at Central Control Room as specified.

Voltage for the voltage regulating circuit shall be obtained from external voltage transformers connected to the terminals of the transformer winding(s).

A safety switch, interlocked with the hand crank, shall be provided so that mechanism cannot be operated by the crank unless all control circuits are open.

A removable handle shall be provided for manual operation of the tap changer. The handle shall automatically block the electrical control when it is in its operating position. Manual operation shall be operable from ground level. Alternatively, a platform and steps may be provided to give access for manual operation.

The tap changer controls and auxiliary equipment shall preferably be mounted in the transformer control cabinet, but may be housed in their own IP 55 cabinet accessible from ground level. The cabinet shall have padlocking facilities. A shatter-proof window shall be installed in the cabinet to permit reading of the tap changer position.

The tap changer control shall consist of the following:

- (1) Mechanical tap position indicator. The indicator shall be visible by a man operating the tap changer manually with the handle.
- (2) Mechanical operations counter.
- (3) “Raise-Lower” push buttons or selector switch, or manual electric control.
- (4) “Manual/Automatic” selector switch.
- (5) “Local/Remote” selector switch.
- (6) Voltage regulating relay.
- (7) Time delay relay, adjustable from 0 to 120 seconds.
- (8) Line drop compensation equipment (where required).
- (9) Parallel control equipment (where required).
- (10) Maximum limit switch.
- (11) Facilities to block operation if the voltage is above or below pre-set values.
- (12) Tap changer motor protection and control circuit protection devices.
- (13) Terminal blocks and test switches.
- (14) Door-actuated internal light.
- (15) Anti-condensation heater with isolation fuses.
- (16) Convenience outlet of approved type.
- (17) Telephone jack.

4.7 Cooling System

4.7.1 General

The cooling system shall consist of the necessary radiators, heat exchangers, fans, pumps and accessories; separately or in such combinations as required to maintain the specified temperature rise and MVA ratings with any one cooling element; i.e. radiator or heat exchanger, out of service. All cooling equipment shall include the necessary supports, mountings, automatic control equipment, conduits and wiring.

All radiators and heat exchangers shall be equipped with bolted flanges and valves to permit the removal of any radiator or heat exchanger without draining the oil from the transformer or any other radiator or heat exchanger. A minimum 38 mm drain valve and plug shall be provided in the bottom of each radiator or heat exchanger. The design and construction of the radiators and heat exchangers shall

eliminate pockets where moisture can accumulate or which can prevent application of a continuous film of paint. Bracing of radiator tubes shall not be by direct welding of the bracing to the tubes, fins and panels.

Both sides of each oil pump shall be equipped with bolted flanges and valves to permit the removal of an oil pump without draining the oil from the transformer.

All fan and pump motors shall be provided with weatherproof flexible leads and threaded, locking type plug-in connectors which are readily accessible for maintenance removal. All fans shall be provided with corrosion-proof guards.

Automatic control equipment for forced-cooled transformers shall have all features specified basically including the following:

- (1) Winding temperature device with electrically independent contacts to control single or double-bank cooling and to initiate remote alarm.
- (2) Switch for automatic and manual control.
- (3) A control cabinet enclosing all control and switching equipment shall be located at the transformer with the top a height not greater than 180 cm above the base. Equipment that requires local operation; i.e. breakers, control switches, alarm test switches etc., shall be located in the lower portion of the cabinet. A drain hole (minimum 15 mm diameter) complete with a screen shall be provided at the low point of the enclosure. The cabinet shall be furnished with thermostatically controlled heaters.

4.7.2 Motors

Motors shall be 400 V, 3-phase, 50 Hz, heavy duty, and squirrel-cage induction type having highest possible efficiency and power factor compatible with good motor design. All motors shall comply with IEC 60034 and shall have dimensions of standard size in accordance with IEC 60072.

Motors shall be rated for continuous operation in a 40°C ambient temperature. The motor winding temperature rise by resistance shall not exceed 80°C.

Motor shall be in accordance with Technical Specification E- 210 Low voltage motor.

4.8 Control Cabinet, Devices and Wiring

4.8.1 General

The transformer control cabinet shall be mounted on the transformer and shall be furnished complete with all devices mounted and wired in the factory before being shipped to the site. If separate cabinets are furnished; such as, the single phase transformer marshalling cabinet or the on-load tap changers remote control cabinets, each cabinet shall be freestanding and shall comply with all other applicable standards.

The Contractor shall provide test switches for all alarm circuits. Test switches shall be wired between alarms and terminal blocks for incoming cables and shall be located inside the control cabinet.

The Contractor shall comply to E-001, E-230, E-231 E-280 ,and all applicable Technical Specification requirement.

4.8.2 Power Supply

Two (2) electric power feeds for operation of the auxiliary equipment shall be furnished by the Contractor. The Contractor shall select and size his electrical equipment accordingly. The Contractor's equipment shall be adequately rated for the short circuit current levels at the supply switchgear and also to meet the specified or expected functional and performance requirements of his equipment.

400 V $\pm$ 10%, 50 kA rms short circuit level, 3 phase, from 400 V MCC's unless otherwise specified. (Current limiting breakers or fuses may be used to limit the short circuit current).

4.8.3 Control and Lighting Devices

Electrical relays shall comply with IEC 60255. The Contractor shall wire all spare contacts from each auxiliary relays, control switch and other devices to terminal blocks. All auxiliary relays shall be furnished in their own enclosures within the cabinet.

Push buttons shall be heavy duty oil-tight type. Switches shall be of the heavy duty type with pistol grip handles or approved equal. The Contractor may substitute push buttons for control or selector switches if approved by the Owner. Number of stages and contacts, number of switch positions and handle style and color shall

be subject to review by the Owner.

Instruments shall comply with IEC 60051. Indicating lights shall be in accordance with IEC 60073 and IEC 60947 and complete with bulb, translucent lens and suitable voltage reducing resistor to increase the bulb life. The 220 Volt AC fluorescent interior lighting with switches, and 3 wire plug receptacles shall be furnished inside each control panel. The light fixtures shall have a degree of protection against electrical shock according to IEC standards and shall be grounded (if required). They shall be connected via suitable fuses to the 230 Volts AC power supply.

4.8.4 Cable Boxes and Glands

Electrical equipment supplied under this Contract shall be fitted with all necessary cable termination boxes which shall be complete with all required fittings. Boxes shall be adequate size to accommodate all cable fittings including end insulation and designed in such a manner that they can be opened for inspection where appropriate without disturbing the gland plate or incoming cable.

Internal clearance between live parts and grounded metal parts shall be adequate to withstand the specified impulse and test voltages, without reliance on an insulating material. All cable boxes and gland plates shall be designed to withstand the high voltage DC cable tests prescribed in IEC 60502 or approved standards.

The Contractor shall comply to E-001, E-230, E-231 E-280 ,and all applicable Technical Specification requirement.

4.8.5 Cables and Wiring

General

- (1) The items wired to local junction boxes shall include power and control components such as flow, pressure and limit switches, temperature devices, etc. Cables and wires shall conform to IEC-60228 and refer to Technical Specification E-230.
- (2) Cables outside of an enclosure or junction box shall be protected by conduit. All multi-conductor cables shall be provided with spiral wound galvanized, steel wire or tape armor. Single conductor cables are not be used.
- (3) Shielding shall be provided on all instrument and thermocouple cables and on control cables whenever required to protect the signals from high voltage

fields or induced voltages.

- (4) The cabinets shall be equipped with a copper ground bus of not less than 7x25 mm cross section extending the entire length of the panel. The ground bus shall be securely anchored to the panel so as to effectively ground the entire structure and shall have provisions, for ground cables at both ends.

4.8.6 Terminal Blocks

Screw type terminals with washer head screws shall be provided for terminating all wiring and interconnecting cables. Barriers between terminals shall be of such height that terminal screws or wire lugs will not project above the barriers. Space between barriers shall be sufficient to accommodate lugs which will not exceed 9.5 mm in width. Terminals shall be clearly and permanently identified.

The Contractor shall comply to E-001, E-230, E-231 E-280 ,and all applicable Technical Specification requirement.

4.8.7 Fuses and Links

Fuse and link holders and bases shall be in accordance with IEC 60269 and 60282, as applicable. Fuse links, fuse contacts and fixed contacts shall be shielded to prevent inadvertent contact with live metal when the fuse is being inserted or withdrawn. Refer to E-001

4.8.8 Motor Control Gear

Control gear shall comply with the requirement of IEC 60947, the control gear being rated according to the duty imposed by the particular application.

Motor contactors shall comply with IEC 60947 class of intermittent duty 0-3 with type 52 enclosure protection and a utilization category AC4. They and their associated apparatus shall be capable of switching the stalled current and shall have a continuous current rating of at least 50 % greater than the full load current of the motors they control.

The operating currents of overload trips fitted to motor contactors shall be substantially independent of ambient temperature conditions, including the effect of direct sunlight on the enclosure in which the contactors are installed.

Where small motors are connected in groups, the group protection shall be arranged so that it will operate satisfactorily in the event of a fault occurring on a

single motor. The control and protection equipment shall be accommodated in the control or marshalling cabinet.

For three-phase motors the protective device shall trip all three phases simultaneously in the event of either a three-phase, phase-to-phase or phase-to-ground fault or overload condition.

4.8.9 Automatic Transfer Switch (If specified)

Automatic transfer switch, if required, shall be 3-pole and of the required ampere rating. It shall be arranged to transfer the load from the normal source of 380 Volts, 3-phase, 50 Hz to the alternate source of 400 volts, 3-phase, 50 Hz when any phase of the normal source drops below 70 % rated voltage, unless otherwise indicated in specification.

The automatic transfer switch shall be a double-throw switch operated by a single coil mechanism momentarily energized. The switch shall be interlocked mechanically and electrically to prevent one contact from closing before the other opens. Operating current for transfer shall be obtained from the source to switch the load is to be transferred. Failure of any coil or disarrangement of any part shall not permit a neutral position. The switch shall be positively locked mechanically on either source without the use of hooks, delicate latches or springs. Gravity or spring-operate switches are not acceptable. All main contact shall be segmented and shall be shielded by glass polyester inter phase barriers. All main contact shall be protected by separate arcing contacts with blowout coils and arc chutes. Switch shall be Automatic Switch Company's Catalog or accepted equal. The transfer switch shall meet requirements of UL 1008.

The transfer switch shall not use current limiting fuses to limit the momentary current that it will be exposed to under fault conditions.

4.9 Photographs

The Contractor shall furnish three complete sets of photographs of each core and coil assembly. The photographs shall be taken just prior to placing the completed core and coil assembly into the tank. A total of five different views shall be provided as follows: Top view, Front view, Left side view, Rear view, Right side view.

4.10 Nameplates

The transformer shall be provided with nameplates in accordance with IEC 60076. The transformer bushing shall be provided with nameplates in accordance with IEC 60137. Name plate shall be as E-001.

The nameplates for rear of instruments and devices shall be engraved with English and Indonesian language legends. Each panel and enclosure shall also be furnished with a nameplate with English and Indonesian languages. A detailed nameplate list with service legends shall be submitted for Owner's review.

The following listed nameplates shall be furnished:

- (1) Owner's transformer identification number plate.
- (2) Rating plate with main windings and current transformer connection diagram.
- (3) Wiring diagram plate showing fan-cooling equipment with accessories with control cabinet terminal-block connection details.
- (4) Valve plate showing basic valve locations, duties and normal operating position.

4.11 Painting

The transformer tank cover and external accessories shall be thoroughly cleaned, primed and painted, in accordance with applicable Civil and Building Technical Specifications S-420 and S-430.

4.12 Shipping

The transformer tank, core and coils, tap changers, radiators, heat exchangers, control equipment, bushing, surge arresters and all accessory equipment shall be adequately anchored, braced and packed to prevent damage from vibration, shock or damages that might reasonably be encountered in transportation and handling.

Transformers shall be shipped with the core and coils immersed in oil unless prohibited by shipping limitations. If core and coils cannot be shipped immersed in oil, transformer tank shall be shipped filled with nitrogen gas or dry air under pressure.

Where transformers are shipped with nitrogen gas or dry air in place of oil, all

valves shall be either plugged or covered to prevent tampering while in transit. A gauge, or a connection for a gauge, shall be provided to indicate the air or gas pressure.

The transformer manufacturer shall place “Warning Signs”, if required, in conspicuous locations on the transformer tank and cover, indicating the danger of asphyxiation if the transformer tank is not purged before entering.

The nitrogen gas and dry air shall be refer to manufacture recommendation

ADD1
-IIE
-004

The Contractor shall attach to each shipment an impact recorder which shall be capable of recording shock in three axes. The impact recorder shall remain the property of the Contractor. The chart and three copies of it shall be delivered to the Owner’s representative at the site.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 General

The Contractor shall perform all type tests, routine tests and special tests necessary to demonstrate the suitability of his equipment. All tests reports and certificates shall be subjected to the approval of the Owner and shall be in accordance with Subsection 5.5 of this specification.

The plant and equipment are subject to inspection by the Owner during the course and on completion of manufacture, factory testing, erection and site testing to ensure compliance with this specification and to provide the necessary operating data.

The Contractor shall supply copies of the relevant test report certificates to the Owner, as soon as practical after each factory test, site test and commissioning.

The subsequent sections list specific inspections, factory tests and site tests which the Owner requires. This shall not preclude the Owner’s right to call for further tests if he considers these necessary to verify the suitability of the equipment

After the plant has passed the site tests required under this bid document and has become available for Commercial Operation, certain additional tests may be carried out in order to investigate the response and recovery of the system during

events such as switching of various items to plant, system faults and load rejection.

The tests shall be arranged to represent working conditions as closely as possible. Where type tests are specified in addition to routine tests, the Contractor may submit type test certificate in lieu of actual type tests provided these have been independently witnessed and cover identical items of plant.

5.2 Factory Tests

5.2.1 Transformer Tests

Routine, type and special tests shall be carried out in accordance with IEC 60076. The tapping connection to be used for the impulse tests shall be the one on which the highest voltage stresses occur. Verification of the correct choice of tapping shall be provided. Impulse tests shall be applied by direct application to each line terminal in turn except where, by agreement with the Owner, the transferred surge method of the test may be adapted for tests on lower voltage windings. The following additional tests shall be made.

(1) Routine Tests

- a. The impedance voltage test should also be carried out on the maximum and minimum tapping positions.
- b. Magnetic circuit insulation (where appropriate).
 - b.1 A power frequency voltage of 2 kV for 1 minute applied as follows :
 - b.1.1 Core bolts to core, to yoke clamps and to core leg side plates.
 - b.1.2 Core to yoke clamps and to core leg side plates.
 - b.2 Immediately prior to dispatch, 2 kV for 1 minute applied between core and earth. A megger may be used for this test.
- c. No load current at :
 - c.1 90% rated voltage and frequency
 - c.2 100% rated voltage and frequency
 - c.3 110% rated voltage and frequency
 - c.4 The maximum voltage equivalent to the value quoted in the specification.
- d. Noise level

e. Vibration

Vibration measurements shall be taken and the level recorder shall be subject to approval. This test shall be carried out unless it can be shown to the satisfaction of the Owner that the level of vibration in the transformer and its auxiliaries is harmless.

f. Partial Discharge (PD)

The details of the PD tests shall be as far as possible based on IEC 60270 subject to the agreement of the Owner except where otherwise stated. In the following specification, where possible, reference will be made to the relevant parts of IEC 60270. Those aspects of the tests not covered by IEC 60270 will be specified herein.

f.1 Test Conditions

The tests shall be made under controlled ambient PD conditions; if possible in a larger shielded enclosure, using corona-free terminals and connections as nearly as possible, except where this may interfere with the achievement of a low ambient PD level.

f.2 Test Procedure

The test circuit, method of measurement and calibration shall be according to IEC 60270.

(2) Type Tests

a. Capacitance

The capacitance between windings and between each winding and earth shall be measured.

b. Switching Surge

Each high voltage terminal of 300 kV and above shall be switching surge tested, with one reduced voltage wave (50%-70%) and three full voltage waves, of negative polarity, between line terminal and ground.

The switching surge waveform shall have the following characteristic.

b.1 A crest value as specified.

b.2 The wave front time (time to crest) shall be $100 \mu\text{s} \pm 20\%$ and the wave tail shall be a minimum of $1000 \mu\text{s}$ and shall exceed 90% of its crest value for at least $200\mu\text{s}$.

c. Transferred Surge

Where recurrent surge oscillograph tests are required, the transformer shall be designed so that with the test voltage applied to the other windings the maximum surge that can be transferred to the unloaded winding(s) does not exceed its specified insulation level. The generator transformer shall also be tested with an external equivalent circuit on the low-voltage winding simulating the load, bus capacitance, surge impedance of the generator units, etc.

During the temperature rise test, the accuracy of oil and/or winding temperature indicating devices shall be determined.

Where impulse test is required, it shall follow the power frequency tests.

Where partial discharge, switching and transferred surge and temperature rise tests have previously been carried out on equipment similar in all essential respects to that included in this Contract, the Owner may waive one or all these tests on production of complete tests records which are considered satisfactory and after the Owner has been credited with the agreed cost.

(3) Special Dielectric Tests

- a. Lightning impulse voltage test shall be applied between each winding terminal and ground in the following order:

1 st -	Reduced full-wave	One application
2nd -	Chopped-wave	Two applications
3rd -	Full-wave	One application

- b. The crest value of voltage for the chopped-wave and full-wave test shall be in accordance with ANSI C 57.12.90. The crest value of voltage for the reduced full-wave test should be between 50 % and 70% of the full-wave test.
- c. A neutral current measurement shall follow each full-wave and reduced full-wave lightning impulse test if current records are not obtained simul-

taneously with voltage records.

- d. A switching impulse test shall be applied between each HV and LV winding terminal and ground. The crest value of the voltage wave shall be in accordance with ANSI C57.12.90.
- e. All windings designed for wye connection shall receive an applied potential to ground test at a voltage level as per ANSI C57.12.90, corresponding to the insulation class at the neutral.
- f. All windings shall receive an induced-potential testing accordance with ANSI C57.12.90. For three-phase transformers, the minimum phase-to-phase induced test voltage shall be as per ANSI C57.12.90.
- g. A corona test shall be made on the transformer. Test shall be made at the following voltage levels and in accordance with the latest NEMA corona test specification. The Radio Influence Voltage (RIV) shall not exceed 500 microvolts when measured at the phase-to-phase test voltages as per ANSI C57.12.90. The Contractor shall use for measurement of the RIV, NEMA Standard 107, "Methods of Measurement of Radio Influence (RIV) of High-Voltage Apparatus", except that the capacitance taps on the high-voltage bushings may be used in place of a separate coupling capacitor.
- h. A description of the proposed corona test, stating the test voltage values at which corona readings will be taken and the maximum guaranteed RIV levels, shall be included in the proposal.

(4) Special Tests and Test Reports

a. Transformer (Heat Run) Temperature Test

A heat-run shall be made at the forced cooled rating until constant temperature is obtained. The test shall be in accordance with ANSI C57.12.90. Heat-run shall be made at the voltage tap which corresponds to the maximum transformer loss at rated power factor while maintaining rated output voltage and current. Sufficient temperature measurements shall be taken to assure that no part of the copper winding, core or structural members exceeds the ANSI allowable temperature rise. The test reports shall show the winding average and hot spot temperature rises, top oil temperature rise and the thermal time constant for each stage of cooling. The test report shall also include the curves and data used for calibration of winding temperature indicators and RTD for winding hot spot

devices. Before and after temperature tests, a dissolved gas analysis shall be performed. All cooling equipment shall be installed on the transformer for the test.

b. Transformer Test Records

Current and voltage oscillograms shall be recorded simultaneously for all impulse tests. The impulse tests shall be recorded in numerical order of performance and oscillograms marked accordingly. The oscillograms shall be recorded on fine grain film and the amplitude of the traces shall be not less than 50 % of film size. The original film records shall be retained by the Contractor and shall be available to the Owner on request. All oscillograph film record shall be identified by the following:

- b.1 Serial number of unit
- b.2 Test number
- b.3 Terminal to which test is applied
- b.4 Type of impulse wave applied
- b.5 Trace identification (current or voltage)
- b.6 Tuning wave
- b.7 Base line

The impulse test oscillographs included in the certified test report shall be 215 mm by 280 mm archival film positives or no erasable Mylar reproducible. Each oscillograph shall be accompanied by a diagram showing the test connection and component values of the circuit.

c. Regulation

Transformer regulation shall be determined for the maximum the forced-cooled rating at unity, 0.9 leading and 0.8 lagging power factors.

5.2.2 Voltage Control Equipment

Routine and type tests shall be carried out in accordance with IEC 60214.

5.2.3 Cable Boxes and Disconnecting Chambers

Oil tightness

All cable boxes and disconnecting chambers shall be tested with oil, having a viscosity not greater than that of IEC 60296 insulating oil when at a temperature of

15°C, at a pressure of 70 kPa (0.7 bar) for 12 hours; during this time no leakage shall occur nor shall there be any permanent set when the pressure is released.

5.2.4 Bushings

Routine, type, sample and special tests shall be carried out in accordance with IEC 60137. A special short time current test shall be made such that each type of bushing being provided shall be subjected to a short time current test in accordance with IEC 62271 for 3 seconds at the fault rating specified.

5.2.5 Tanks and ONAN Coolers

(1) Routine Tests

a. Oil Leakage

All tank and oil filled compartments including all forms or radiators but excluding separate coolers using forced oil circulation, (see Clause 5.2.6) shall be tested, before painting, for oil tightness by being completely filled with oil of viscosity not greater than that of IEC 60296 insulating oil at a temperature of 15° C and subjected to a pressure equal to the normal pressure +35 kPa (0.35 bar). This pressure shall be maintained for a period of not less than 24 hours, during which time no leakage shall occur.

b. The tap changer barrier shall be subjected to normal oil pressure head for 24 hours, during which time there shall be no leakage from the panel or bushings.

c. Detachable radiators may be tested as separate units.

(1) Type Tests

a. Vacuum

One (1) transformer tank, tap changing compartment, radiator and cooler of each size shall be subjected, when empty of oil, to that vacuum test level specified. There shall be no permanent deflection of the stiffeners, nor shall the deflection of the panels exceed the value specified in the following table.

Major dimension of panel between stiffeners and vertical or horizontal (meters)	Maximum deflection
Up to 1.5 m	3 mm
Above 1.5 m up to 3 m	8 mm
Above 3 m	13 mm

b. Pressure

One (1) transformer tank tap changing compartment, radiator and cooler of each size shall be subjected to a pressure corresponding to the normal pressure + 35 kPa (0.35 bar) for 12 hours. There shall be no permanent deflection of the stiffeners nor shall the permanent deflection of panels between stiffeners exceed the value specified in the above table. This test may be combined with a routine oil leakage test.

The tap changer barrier shall be shown to withstand an over pressure test of normal pressure + 35 kPa (0.35 bar) for 12 hours.

c. Pressure Relief Device

A pressure relief device of each size shall be subjected to increasing oil pressure and shall operate before reaching normal pressure 0.35 bar. The operating pressure shall be recorded on the test certificate.

5.2.6 Cooling Plant with Forced Oil Circulation

(1) Routine Tests

All air/oil coolers using forced oil circulation shall be filled with oil of a viscosity not greater than that of IEC 60296 insulating oil at a temperature of 15°C and subjected to a pressure equal to twice the maximum working pressure.

(2) Type Tests

One (1) forced-oil unit of each type shall be subjected, when empty of oil, to that vacuum test level specified in the relevant standards.

5.2.7 Pump, Motors, Pipework, Oil Sampling Devices and Valves

(1) Routine Tests

a. Oil filled Equipment

The bodies of all oil pumps complete with submerged motors, if any and the oil pipeworks, oil sampling device and valves shall withstand a hydraulic pressure of 140 kPa (1.4 bar) for 15 minutes.

b. Motors

Each machine shall be subjected to the following tests where applicable:

b.1 Measurements of winding resistance (cold).

b.2 No load test at rated voltage for determination of fixed losses.

c. An overvoltage test at 1.5 times rated voltage applied with the machine running at no load, for a period of 3 minutes, to test interturn insulation.

d. High voltage test in accordance with IEC 60034.

(2) Type Tests

a. Motors

Performance tests shall be in accordance with IEC 60034, however, certificates of type tests in accordance with IEC will be accepted.

5.2.8 Oil Tests

Samples of oil from each consignment shall be tested in accordance with IEC 60296 before dispatched. Before commissioning any transformer, the dielectric strength of its oil shall be checked, tested and the results approved by the Owner.

5.2.9 Gas and Oil Actuated Relays

The tests shall be made on relays when completely assembled according to IEC. Where oil is referred to it shall have a viscosity not greater than that of IEC 60296 insulating oil at 15° C.

5.2.10 Instrument Transformers

Production tests for instrument transformer shall be in accordance with IEC 60044 and 60186. A test certificate and excitation curve shall be provided for each type of transformer. The current transformer excitation, ratio and phase angle curves shall be verified as part of the current transformer routine tests.

5.2.11 Insulation System

Type tests for the bus bar and insulator systems of standard assemblies shall demonstrate the system's ability to withstand rated short circuit current for one (1) second.

Type tests shall demonstrate the ability of the insulation systems proposed to withstand the climate at the site without bacterial or fungus attack or other deterioration.

5.2.12 Control Cabinets, Devices and Wiring

- (1) All enclosures shall be type tested in accordance with IEC 60529.
- (2) Each lot of conduits shall be tested in accordance with the applicable section of IEC 60614.
- (3) Power cables shall meet the test requirements of IEC 60502 and 60811.
- (4) Relays shall be type tested to demonstrate that they can operate successfully under the site conditions. Type and routine tests shall include all tests called for by IEC 60255.
- (5) Point-to-point continuity tests and electrical insulation tests, in accordance with previous Clauses (9) and (10), and applicable IEC or ANSI standards.

5.3 Site Tests

After the plant and ancillary equipment have been erected and connected up on site, the Contactor shall carry out, to the satisfaction of the Owner, such tests as may be required to prove compliance with the specification, independently of any tests carried out at the manufacturer's works.

Not less than thirteen (13) weeks before any section of the plant is required to enter commercial service, the Contractor shall submit for the approval of the Owner his detailed site test proposals. A record of the result of the test in this category shall be made available to the Owner.

The site tests, as a minimum, shall include those tests described in IEC standard

5.4 Inspection and Witnessing of Tests

The Owner shall have the right at all reasonable times to inspect the work and ob-

serve production tests and any other test specified. The Contractor shall make all necessary arrangements and provide all reasonable facilities and access for inspections and observation of such tests.

5.5 Test Reports

The Contractor shall furnish the Owner with certified test reports covering the guaranteed requirements and all test performed or necessary to meet the specified requirements of the equipment.

6. QUALITY ASSURANCE / QUALITY CONTROL

6.1 Requirement

The Contractor shall fulfill the quality assurance/quality control requirements listed for individual equipment in the respective equipment specifications and in Part 4 Section 2 Subsection 2.7.

6.2 Procedures

The quality assurance /quality control procedures shall also cover the following:

- (1) Design criteria and engineering calculations.
- (2) Material control.
- (3) Factory and site inspections.
- (4) Wiring insulation tests.
- (5) Insulation high voltage tests.
- (6) Control and functional tests.
- (7) Protective relay and device settings and tests.
- (8) Welding.
- (9) Nondestructive tests.
- (10) Cleaning and painting.
- (11) Packaging and shipping.

6.3 Documentation

Quality assurance/quality control documentation shall include, but not limited to, the following:

- (1) Inspection Reports.
- (2) Certified Test Reports.
- (3) Material Certifications.
- (4) All other documentation required by applicable codes and standards.

(Last page of E-301)